

Ishan Singh

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Summary

Backend / Platform Engineer with 6+ years of experience building distributed systems, multi-tenant SaaS platforms, IAM, real-time data pipelines, and Kubernetes-native backend services at scale.

Technical Skills

Languages: Go, Python, Java, C/C++, SQL

Backend / Infra: Kubernetes, AWS, Kafka, PostgreSQL, ClickHouse, REST, GraphQL

Security / Observability: IAM, SSO (SAML, OIDC), Splunk, CloudWatch

AI / Infra: AWS Bedrock, MCP, LLMs, LangChain, RAG, Qdrant, pgvector

Experience

Cisco

Software Engineer

Bengaluru, KA

Oct. 2024 – Present

- Designed and built **tenant-scale global search** for Cisco Secure Access using Go, OpenSearch, DynamoDB, and Kafka, indexing 30+ security configuration object types across 50+ searchable fields and millions of cross-object relationships.
- Enabled nested and where-used lookups to trace related objects, policies, and dependencies – targeting **60% lower policy-troubleshooting MTTR** and **40% fewer audit misconfigurations**.
- Designed anomaly schemas, RCA service APIs (Go), and deployment pipelines (Terraform, Kubernetes) for an internal **root-cause analysis platform**; shipped MCP servers in Python with Bedrock-powered agents.
- Led implementation of a distributed firewall log ingestion and enrichment pipeline processing **10K events/sec** using Go, Kafka, Kinesis, DynamoDB, and S3 with no upstream lag.

Deepfactor, acquired by Cisco

Senior Software Engineer

Bengaluru, KA

Feb. 2023 – Oct. 2024

- Designed and built the **IAM layer** using Keycloak and Postgres for Deepfactor SaaS beta, supporting multi-tenant authentication and authorization across SaaS and on-prem deployments.
- Implemented generic **enterprise SSO** support using SAML and OIDC, enabling customer-managed identity provider integrations.
- Built a **rate-limiting control plane** that read tenant usage from ClickHouse and applied Istio configuration for per-tenant enforcement.
- Built team management and **role-scoped access controls** for secure multi-tenant collaboration.

Amazon

Software Development Engineer

Bengaluru, KA

Jul. 2021 – Jan. 2023

- Led organization-wide **platform migration automation** in Java for **50+ teams**, reducing a multi-day manual process to a minutes-long workflow.
- Contributed to **AIWorkbench** (Java), a no-code workflow platform for composing chained workflows across AWS and internal services.
- Designed and implemented **approval workflows** using AWS Lambda and Step Functions to pause execution, validate intermediate ML inference results, and resume downstream processing.
- Built a **one-click campaign launcher** in Python using ML inference outputs to recommend campaign configurations during large-scale marketplace events.
- Diligently handled **production on-call** duties, triaging and resolving customer-impacting issues while improving **runbooks and operational documentation**.
- Raised engineering standards through **peer code reviews**, service documentation, and **mentoring new hires** through onboarding and service ownership.

Optum, UnitedHealth Group

Software Engineer

Remote

Aug. 2020 – Jun. 2021

- Built a service in Java to validate **Kafka streams** against source databases and generate automated data quality reports.
- Migrated legacy services from **Java to Go** and added observability using Prometheus, Grafana, and Jaeger.

Mindship

Software Engineer Intern

Bengaluru, KA

Jun. 2019 – Sep. 2019

- Built **reusable backend services** in Go including a GraphQL-based web chat service and messaging integrations with Twilio and SendGrid.

Projects

Build Your Own Systems: Portable AI coaching skill and systems curriculum with 365 runnable exercises across core CS and infrastructure topics.

Open Agent Policy: Zero-trust access control framework for AI agents with a Python SDK and a policy engine written in Go.

Distributed WAL: Disk-backed distributed write-ahead log in Go with Raft-based replication, leader election, and fault-tolerant recovery.

Lox Interpreter / C++ VM: Tree-walking interpreter in Go and bytecode virtual machine in C++, featuring lexical analysis, recursive-descent parsing, and stack-based execution.

Virtual Stylus: Hand-gesture-driven drawing interface built in Python with real-time object detection and classification for touchless HCI.

Education

Zakir Husain College of Engineering & Technology, AMU

Bachelor of Computer Engineering; GPA: 9.12/10

Aligarh, UP

Aug. 2016 – May 2020